

TEAM DATA MESH

Multi-Modal Graph RAG

A Document Intelligence Platform

Ask questions across all your files — PDFs, images, audio, videos — and get intelligent, sourced answers in real time.

FastAPI · ChromaDB · Neo4j · Groq AI · React · Docker

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Chat
Upload
Search
Visualize

Why Existing Tools Fall Short

Most AI assistants only understand text. Real-world knowledge lives in many formats.

PDFs & Documents

✓ **Traditional:** Can read

✓ **This System:** Can read

Images & Photos

✗ **Traditional:** Ignored

✓ **This System:** AI vision understands

Audio Files

✗ **Traditional:** Ignored

✓ **This System:** Transcribed & searchable

Videos

✗ **Traditional:** Ignored

✓ **This System:** Extracted & indexed

Cross-File Links

✗ **Traditional:** No memory

✓ **This System:** Knowledge graph connects all

The gap: Old tools answer from one file at a time. This system answers from all your files — connected together.

What We Built

File Intelligence

- All formats: PDF, TXT, JPG, PNG, MP3, WAV, MP4, MOV
- AI vision captions every image automatically
- Whisper AI transcribes audio & video
- SHA-256 dedup — no duplicate indexing
- 50 MB limit with clear error feedback

Smart Chat

- Ask in plain English — typos handled
- Streaming responses, word by word
- Clickable source badges under every answer
- Context-aware follow-ups across turns
- Open images, play audio, view PDFs inline

What We Built continued

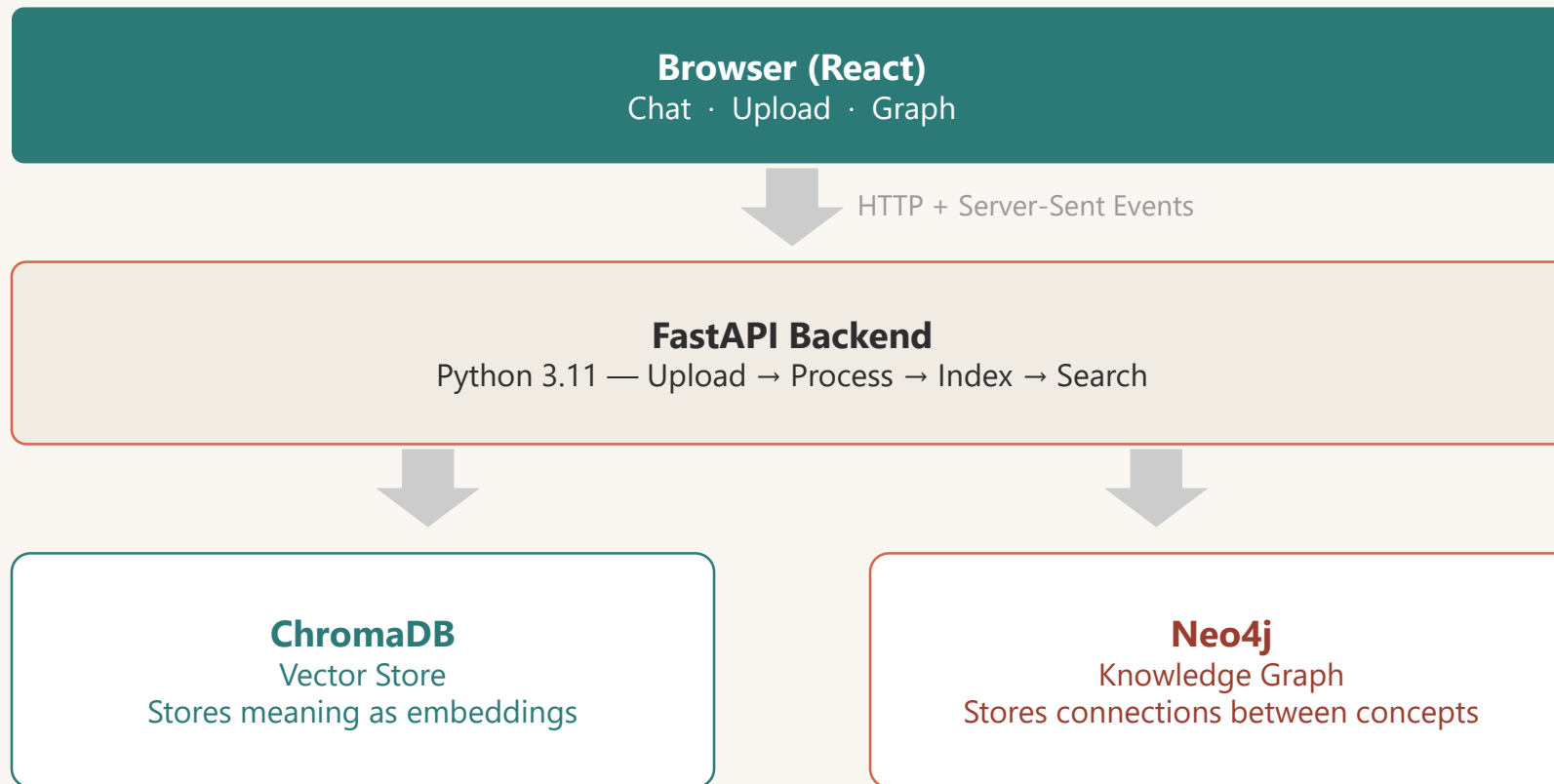
Knowledge Graph

- Extracts entities: people, places, concepts, tech
- Maps relationships across documents
- Auto-heals on file deletion using LLM
- Interactive D3.js visualization in browser

Engineering

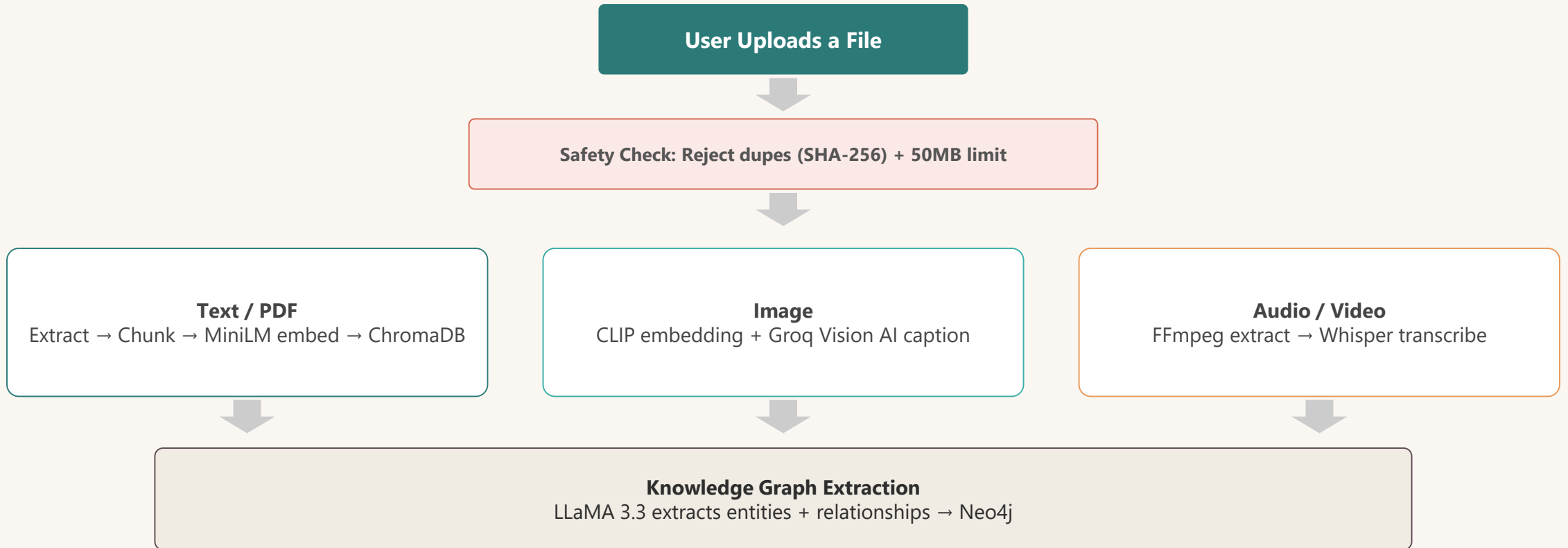
- Fully Dockerized — 4 services, one command
- Parallel uploads — 10 files simultaneously
- Server-Sent Events for streaming
- Production-grade error handling

System Architecture



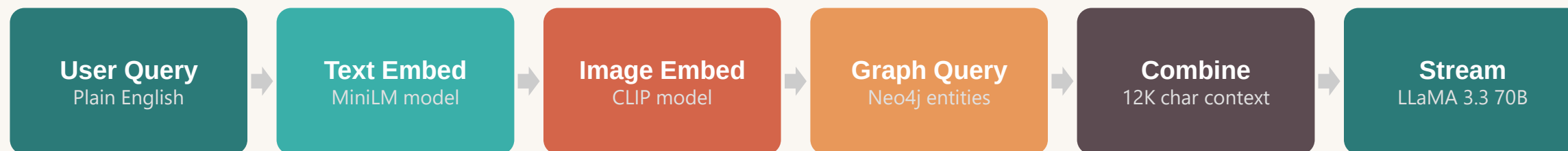
ChromaDB finds relevant content. **Neo4j** finds connected knowledge.
Together they produce richer, more accurate answers than either alone.

File Processing Pipeline



Every file — regardless of type — becomes searchable text and graph knowledge within seconds.

How the AI Answers



The Search Process

- Text + image embeddings search ChromaDB in parallel
- Keywords extract entities from Neo4j knowledge graph
- All context combined and capped at 12,000 characters
- LLaMA 3.3 70B on Groq streams answer token by token
- Clickable source badges link back to original files

Smart Retrieval

Typo Forgiveness

User: *"show me vdios"*

→ Videos returned

User: *"giv me aduio"*

→ Audio returned

difflib fuzzy match at 0.65 cutoff

Context Follow-ups

User: *"show me city.jpg"*

→ Image shown

User: *"tell me about it"*

→ Describes city.jpg

Scans last 4 conversation turns

Filename Search

User: *"bitcoin"*

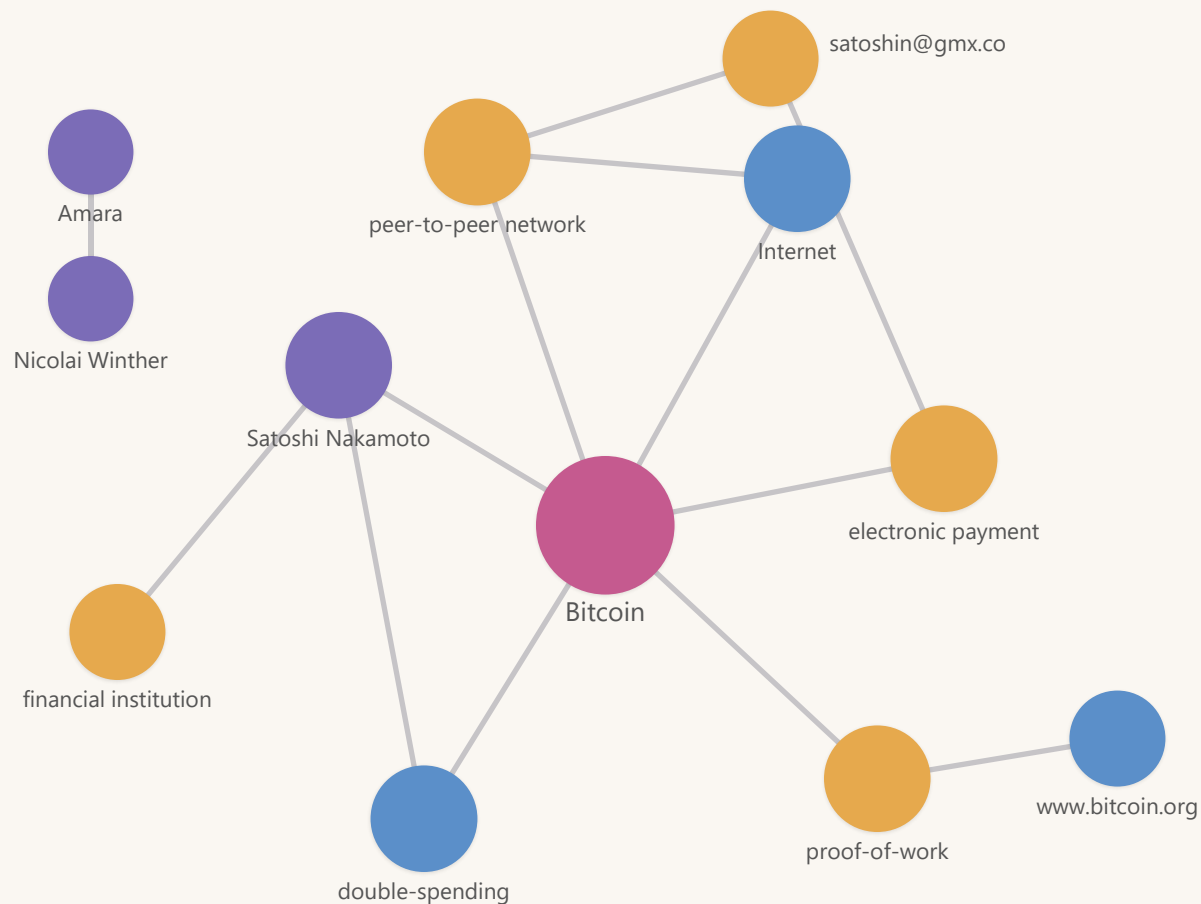
→ bitcoin_whitepaper.pdf

User: *"city"*

→ city.jpg

Partial match on any filename

Knowledge Graph



How It Works

- Every file adds entities and edges to Neo4j
- Cross-document connections emerge automatically
- Isolated clusters show unrelated documents

Auto-Heal on Delete

- 1 Remove vectors + graph entities
- 2 Find disconnected neighbors
- 3 LLaMA evaluates if direct link makes sense
- 4 Auto-create link — graph stays intact

Tech Stack

All Free & Open Source

Layer	Technology	Purpose
Frontend	React 18 + Vite + Tailwind	Chat, file manager, graph UI
Backend	FastAPI (Python 3.11)	API, processing, orchestration
Vector DB	ChromaDB	Semantic similarity search
Graph DB	Neo4j 5.15	Entity relationship storage
LLM	Groq — LLaMA 3.3 70B	Chat responses, entity extraction
Vision AI	Groq — LLaMA 4 Scout	Image captioning
Speech AI	Groq — Whisper Large v3	Audio/video transcription
Embeddings	MiniLM + CLIP ViT-B/32	Text + image vectors
Deploy	Docker + Docker Compose	One-command deployment

```
$ cp .env.example .env && ./start.sh # starts all 4 containers → localhost:3000
```

Live Demo Screenshots

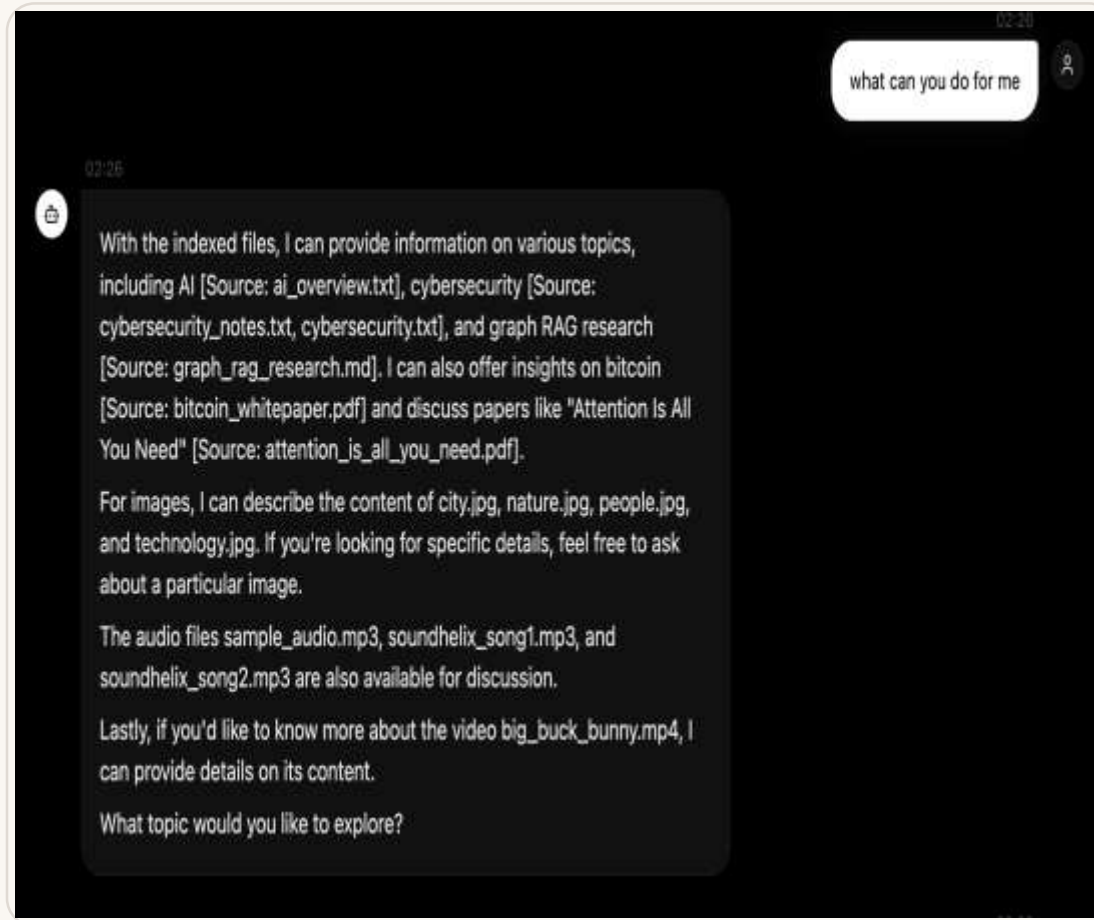
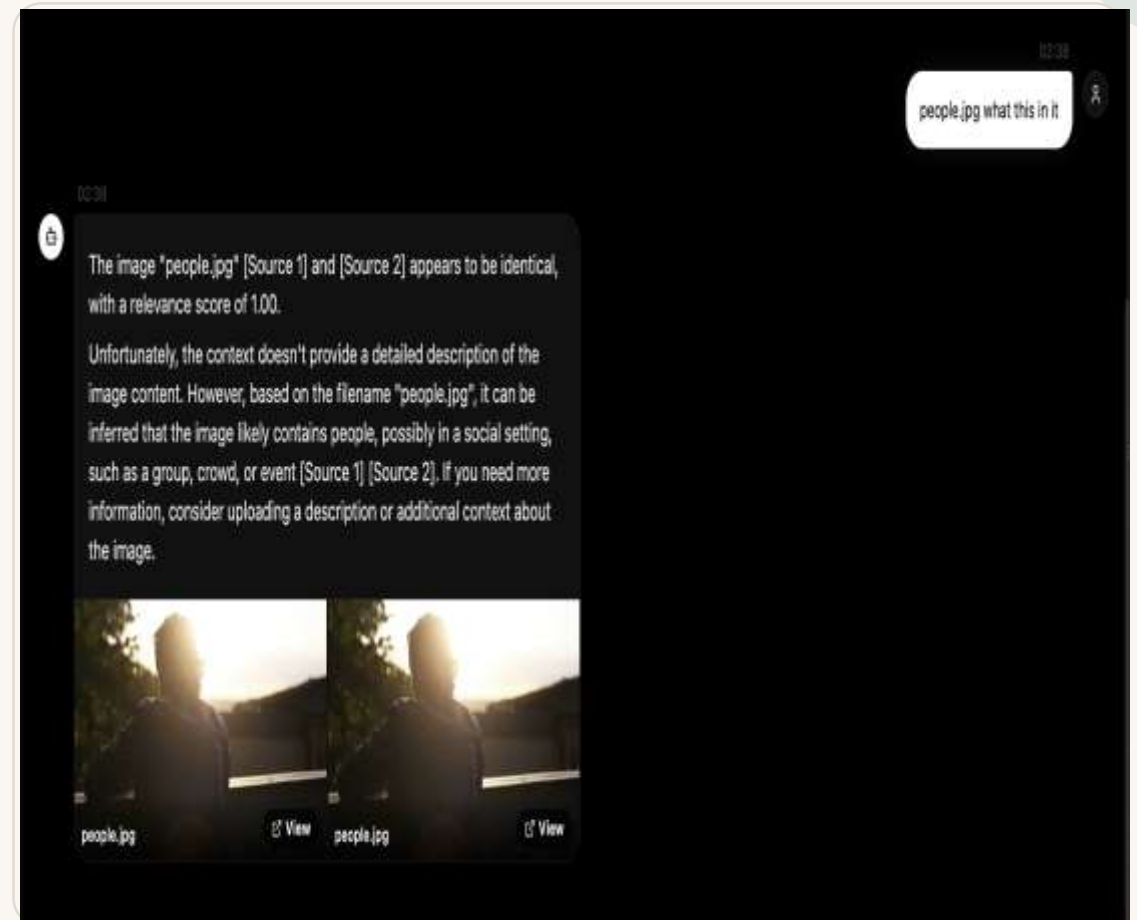


Image Search



Context Follow-ups

Multi-modal search across PDFs, images, audio, and video — with streaming AI responses, clickable source citations, and natural follow-up questions

Results & Takeaways

Working Features

- ✓ 4 file modalities: text, image, audio, video
- ✓ Streaming AI chat under 2 seconds
- ✓ Typo-tolerant fuzzy matching
- ✓ Context-aware follow-up questions
- ✓ Knowledge graph with auto-heal
- ✓ Parallel file uploads
- ✓ One-command Docker deployment

Scale

- Hundreds of files across all modalities
- Thousands of vector chunks in ChromaDB
- Thousands of nodes and edges in Neo4j

Not just a chatbot. A complete document intelligence system — built from scratch — that understands every file type, connects knowledge across documents, and runs entirely on free APIs.

Literature Survey

RAG & Retrieval Systems

- Lewis et al. (2020) — RAG: Retrieval-Augmented Generation for knowledge-intensive NLP tasks
- Gao et al. (2023) — Survey of RAG for large language models, retrieval strategies & fusion
- Borgeaud et al. (2022) — RETRO: Retrieval-enhanced transformers with 2T token database
- Izacard & Grave (2021) — FiD: Fusion-in-Decoder for open-domain QA with multi-passage retrieval
- Karpukhin et al. (2020) — DPR: Dense Passage Retrieval for open-domain question answering

Graph & Multi-Modal AI

- Edge et al. (2024) — GraphRAG: Graph-based retrieval augmented generation from Microsoft
- Yasunaga et al. (2022) — QA-GNN: Knowledge graph reasoning for question answering
- Li et al. (2023) — Multi-modal RAG combining text, image & table understanding
- Radford et al. (2023) — Whisper: Robust speech recognition via large-scale weak supervision
- Liu et al. (2023) — LLaVA: Visual instruction tuning for multi-modal LLMs

References

Cited Works

1. P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in Proc. NeurIPS, 2020, pp. 9459-9474.
2. Y. Gao et al., "Retrieval-Augmented Generation for Large Language Models: A Survey," arXiv:2312.10997, 2023.
3. S. Borgeaud et al., "Improving Language Models by Retrieving from Trillions of Tokens," in Proc. ICML, 2022, pp. 2206-2240.
4. G. Izacard and E. Grave, "Leveraging Passage Retrieval with Generative Models for Open Domain QA," in Proc. EACL, 2021, pp. 874-880.
5. V. Karpukhin et al., "Dense Passage Retrieval for Open-Domain Question Answering," in Proc. EMNLP, 2020, pp. 6769-6781.
6. D. Edge et al., "From Local to Global: A Graph RAG Approach to Query-Focused Summarization," arXiv:2404.16130, 2024.
7. M. Yasunaga et al., "QA-GNN: Reasoning with Language Models and Knowledge Graphs for QA," in Proc. NAACL, 2021, pp. 535-546.
8. J. Li et al., "Multi-Modal Retrieval Augmented Generation for Knowledge-Intensive Visual QA," arXiv:2312.05253, 2023.
9. A. Radford et al., "Robust Speech Recognition via Large-Scale Weak Supervision," in Proc. ICML, 2023, pp. 28492-28518.
10. H. Liu et al., "Visual Instruction Tuning," in Proc. NeurIPS, 2023.